

Recall from last class:

- **Theorem:** Let $\mathcal{B} = \{b_1, \dots, b_n\}$ and $\mathcal{C} = \{c_1, \dots, c_n\}$ be bases of a vector space V . Then there exists a unique $n \times n$ matrix P such that $[x]_{\mathcal{C}} = P[x]_{\mathcal{B}}$.
- We call P the **change of coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$** . In general

$$P = [[b_1]_{\mathcal{C}} \ \dots \ [b_n]_{\mathcal{C}}].$$

Also P^{-1} is the change of coordinates matrix from $\mathcal{C}$ to $\mathcal{B}$.

- Let V be a vector space with basis $\mathcal{B} = \{b_1, \dots, b_n\}$ and W be a vector space with basis $\mathcal{C} = \{c_1, \dots, c_m\}$. Given a linear transformation $T: V \rightarrow W$, we naturally seek an associated matrix to represent T using the bases; namely we want an $m \times n$ matrix M such that $[T(x)]_{\mathcal{C}} = M[x]_{\mathcal{B}}$ for all $x \in V$. We call M the **matrix of T relative to bases $\mathcal{B}$ and $\mathcal{C}$** . In general $M = [[T(b_1)]_{\mathcal{C}} \ \dots \ [T(b_n)]_{\mathcal{C}}]$.
- **Diagonal Representation Theorem:** Suppose $A = PDP^{-1}$ where D is a diagonal matrix and $T(x) = Ax$. If $\mathcal{B}$ is the basis for $\mathbb{R}^n$ formed by the columns of P , then D is the matrix of T relative to $\mathcal{B}$.
- If A and B are $n \times n$ matrices, then A is **similar** to B if there exists an invertible matrix P such that $A = PBP^{-1}$.

Definition 1. If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a non-zero vector v , called an **eigenvector** corresponding to λ , such that

$$Av = \lambda v.$$

1. Let $A = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Determine if v is an eigenvector of A and find the corresponding eigenvalue.

Definition 2. If λ is an eigenvalue of an $n \times n$ matrix A , then the set of all solutions v to $Av = \lambda v$ is the **eigenspace** of A corresponding to λ .

2. Let $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$, then $\lambda = 2$ is an eigenvalue. Find the corresponding eigenspace.

Theorem 1: The eigenvalues of a triangular matrix are the diagonal entries.

3. Prove the theorem.

Theorem 2: If $v_1, \dots, v_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_p$ of an $n \times n$ matrix A , then $v_1, \dots, v_p$ are linearly independent.

4. Prove the theorem

5. For a general $n \times n$ matrix A , how can we find the eigenvalues?

6. Let $A = \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$ Find the eigenvalues of A and corresponding eigenspaces.

7. Prove that λ is an eigenvalue of A if and only if λ is an eigenvalue for A^T .

Definition 3. If A is an $n \times n$ matrix, then the characteristic polynomial of A , denoted by $p_A(\lambda)$, is given by

$$p_A(\lambda) = \det(A - \lambda I_n).$$

Note: λ is an eigenvalue of A if and only if λ is a zero of the characteristic polynomial.

8. Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 6 \\ 0 & 0 & 5 & 5 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ Find the characteristic polynomial of A .

9. If A and B are similar matrices, then they have the same characteristic polynomials and hence the same eigenvalues (with the same algebraic multiplicities).

10. True or False.

- (a) If two matrices have the same eigenvalues, then they are similar.
- (b) If two matrices are row equivalent, then they have the same eigenvalues.
- (c) If two matrices are row equivalent, then they are similar.